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CPCS-214 Computer Architecture and Organization

05 - Technologies for Building Processors and Memory

Introduction

- Processors and memory have improved at an incredible rate
- Because computer designers have embraced the latest in electronic technology to try to win the race to design a better computer

Transistor and VLSI

- An on/off switch controlled by electricity
- *Integrated circuit* (IC) combined dozens to hundreds of transistors into a single chip
- Very Large-Ssale integrated (VLSI) circuit
  – A device containing hundreds of thousands to millions of transistors
- Increased capacity and performance
- Reduced cost

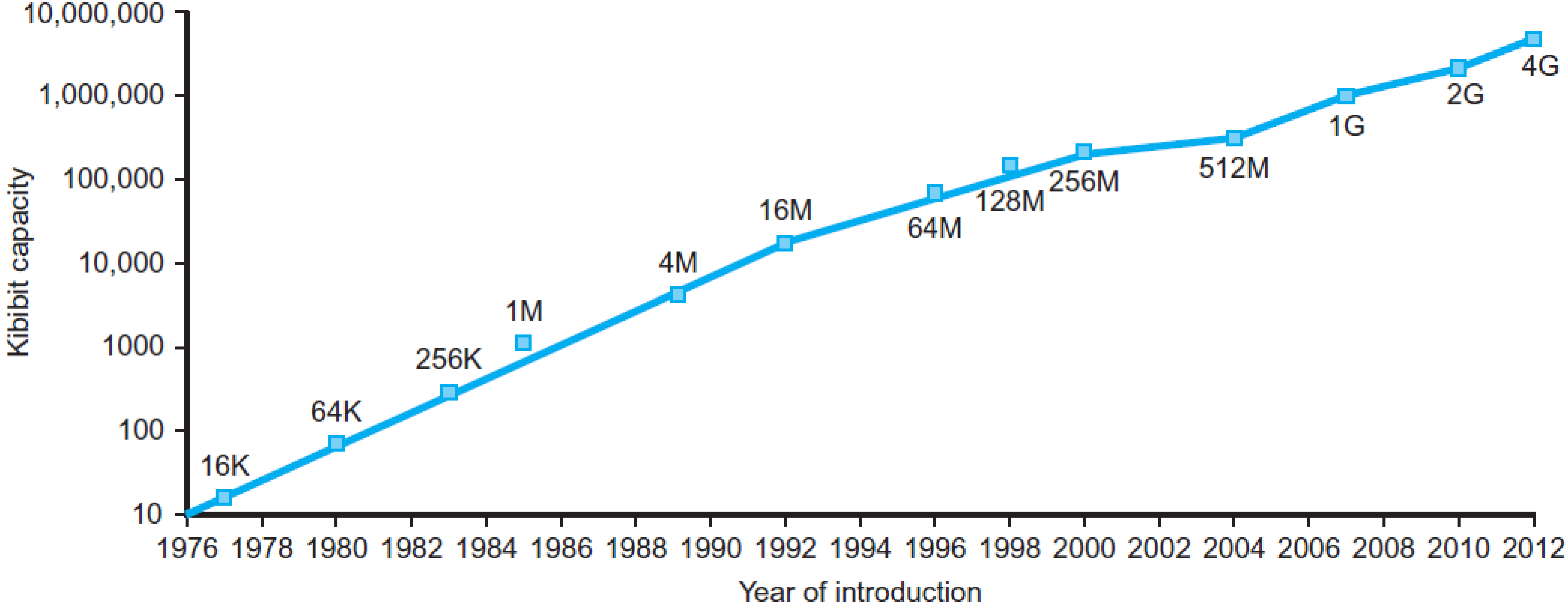
Relative Performance of Technologies

Year	Technology used in computers	Relative performance/unit cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit	900
1995	Very large-scale integrated circuit	2,400,000
2013	Ultra large-scale integrated circuit	250,000,000,000

Capacity per DRAM Chip

- Growth in DRAM capacity
- For decades, industry has consistently quadrupled capacity every 3 years

Capacity per DRAM Chip



Chip Manufacturing Process

- Begins with silicon
  - Substance found in sand
- Because silicon does not conduct electricity well, it is called a semiconductor
- Process starts with a silicon crystal ingot that is between 8 and 12 inches in diameter and about 12 to 24 inches long

Wafer

- Ingot is sliced into wafers no more than 0.1 inches thick
- Wafers go through a series of processing steps, during which patterns of chemicals are placed on each wafer, creating transistors, conductors, and insulators

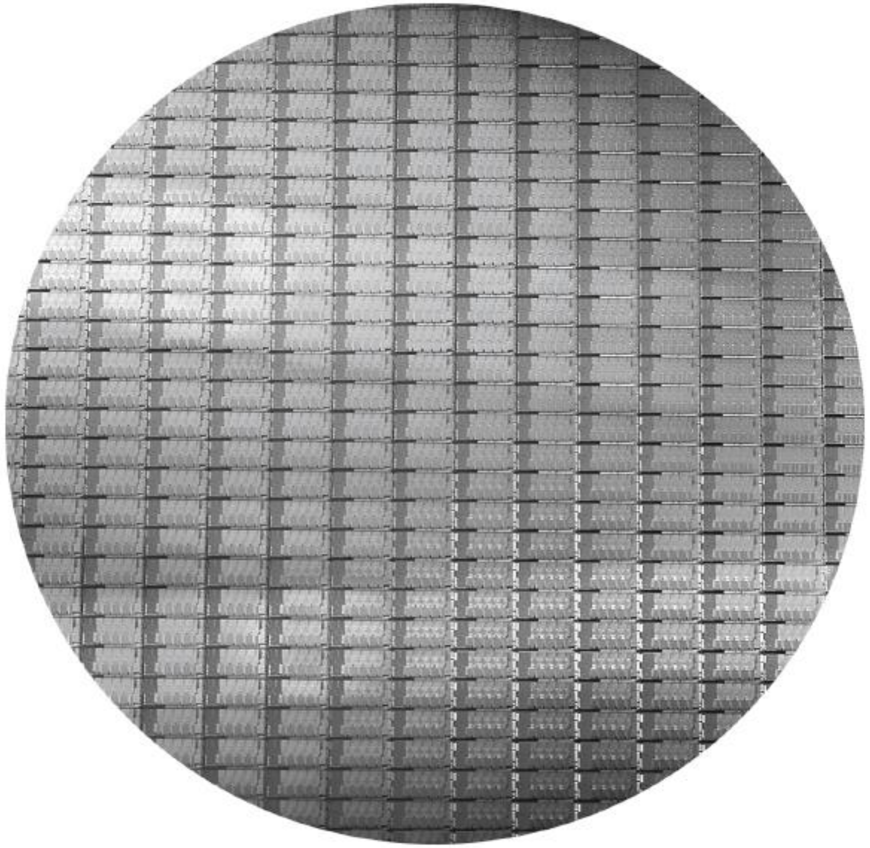
Defects

- Single microscopic flaw in wafer itself or in one of the dozens of patterning steps can result in that area of the wafer failing
- Make it virtually impossible to manufacture a perfect wafer

Dicing

- Patterned wafer is then chopped up, or *diced,* into components called dies and more informally known as chips
- Enables to discard only those dies that were unlucky enough to contain the flaws, rather than the whole wafer

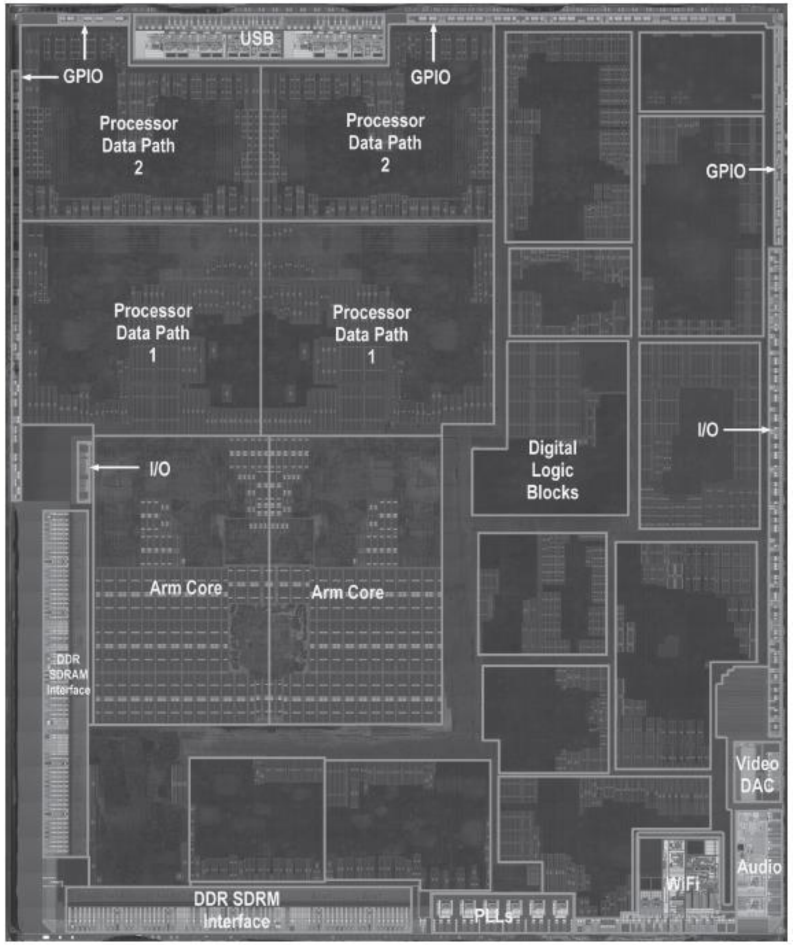
Intel Core i7 Wafer



300mm wafer, 280 chips, 32nm technology

Each chip is 20.7 x 10.5 mm

Individual Microprocessor Die



Yield

- Percentage of good dies from total number of dies on wafer
- Cost of IC rises quickly as die size increases, due both lower yield and the smaller number of dies that fit on a wafer
- To reduce cost, using next generation process shrinks large die as it uses smaller sizes for both transistors and wires
  – Improves yield and die count per wafer
- A 32-nanometer (nm) process was typical in 2012
  – means that the smallest feature size on die is 32 nm

Bonding

- Once found good dies are found, they are connected to the input/output pins of a package, using a process called *bonding*
- Packaged parts are tested a final time, since mistakes can occur in packaging
- Then they are shipped to customers

Chip Manufacturing Process